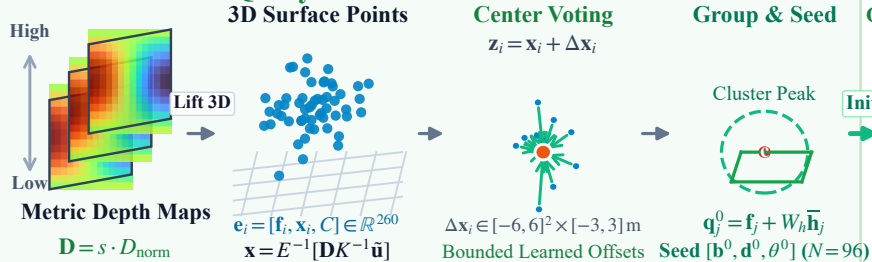


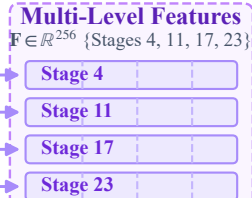
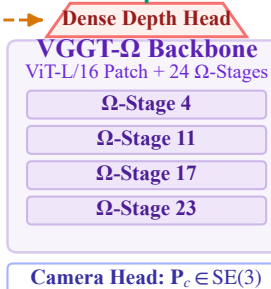
Surface-Guided Query Initialization



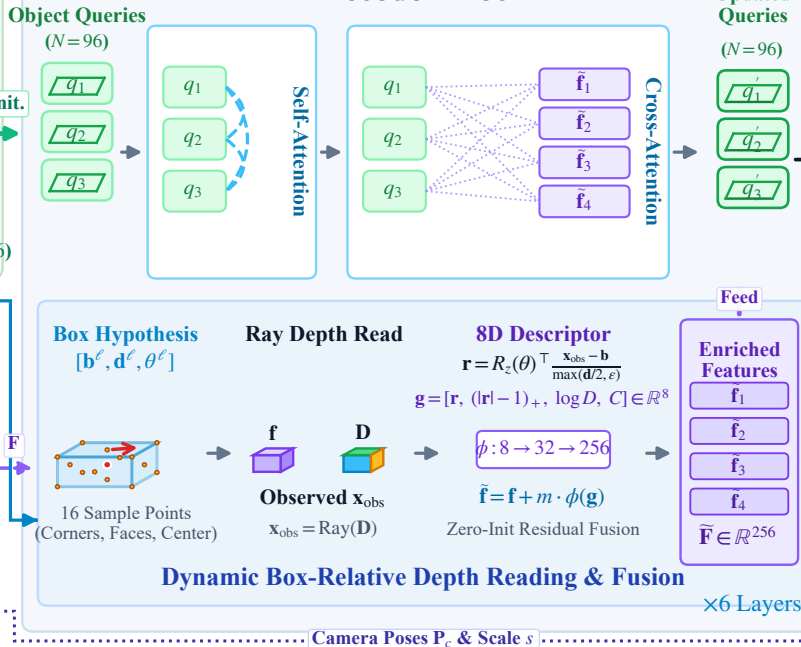
Multi-View Surround
Synchronized (Ego & CAVs)

Rig Metric Scale Solver

$$s = \text{med}\left(\frac{d_{\text{tri}}}{D_{\text{norm}}}\right) \Rightarrow \mathbf{D} = s \cdot \mathbf{D}_{\text{norm}}$$



Decoder Block



Cooperative 3D Detection

Rig Pose Fusion

$$\hat{T}_v = f(\{P_c\}, s)$$

$$\hat{T}_{e \leftarrow v} = \hat{T}_{v_0}^{-1} \hat{T}_{v_j}$$

$$\mathbf{b}_{\text{ego}} = \hat{T}_{e \leftarrow v} \mathbf{b}$$

Rotated BEV NMS

IoU Thresh: 0.15
Score > 0.2

Fused 3D Scene

Occluded Target
(CAV-Aided Discovery)

